AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM129
Technologies in practice

Makeup-Midterm Examination (MTA)
Fall Semester 2025/2026

Date: 03/Dec./2025

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 2h

AOU EGYPT

Instructions:

- Total Marks: **60**
- This exam consists of **6** Questions.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

CamScanner

Question(1) Answer the following statements with True or False and (10 Marks):

1. Volunteered data is captured by recording the actions of individuals, such as location data when using cell phones
2. In computing, the term bit is an abbreviation of "binary digit" and represents the smallest piece of data.
3. Bandwidth measured in Gigabits per second (Gbps) is equivalent to 1012 bits per second (bps)
4. The physical address (MAC address) of a device on an Ethernet network is burned into the Network Interface Card (NIC) and does not change
5. In a hierarchical network design, the Distribution Layer provides the high-speed connection between distribution layer devices.
6. When determining the network portion of an IPv4 address using the subnet mask, the 0s in the subnet mask represent the network portion, and the 1s represent the host portion.
7. Transmission Control Protocol (TCP) is preferred over User Datagram Protocol (UDP) for time-sensitive applications like Voice over IP (VoIP) because TCP's flow control mechanisms minimize latency
8. Disabling the SSID broadcast on a wireless network is sufficient security to protect the WLAN from experienced threat actors
9. The Internet Protocol (IP) layer in the TCP/IP model is responsible for guaranteeing the reliable delivery of information and managing flow control between end devices.
10. In computer architecture, a process running in User mode can execute all instructions, including those designated as privileged instructions

Question (2) Answer the following questions (10 Marks): -

1. Describe the three layers of a hierarchical network design and state the core function of each layer. (3 Marks)
2. Explain the difference between a physical address (MAC address) and a logical address (IP address) (4 Marks)
3. How does the logical AND operation, in conjunction with the subnet mask, enable a host to determine the network portion of an IPv4 address? (3 Marks)

Question (3) Answer the following questions (10 Marks): -

1. Identify the three primary types of media used in modern networks and describe the encoding method for data transmission in each. (4 Marks)
2. Discuss the primary environmental and electrical interference issues associated with twisted-pair copper cable, specifically mentioning UTP and STP cables. (3 Marks)
3. Describe the purpose and core functionality of the ping utility and the traceroute utility (or tracert in Windows) in network verification, noting what information traceroute provides that ping does not. (3 Marks)

Question (4) Answer the following questions (10 Marks): -

1. Compare and contrast the reliability and delivery mechanisms of Transmission Control Protocol (TCP) and User Datagram Protocol (UDP), and specify the type of application where UDP is preferable. (4 Marks)
2. Explain the purpose of port numbers in network communication and define the three categories of port numbers as assigned by ICANN. (3 Marks)
3. Describe how the Domain Name System (DNS) functions when a client tries to access a web server, including the specific port used for DNS queries. (3 Marks)

CamScanner

Question (5) Answer the following questions (10 Marks): -

1. Describe the concept of Dual-Mode Operation in an Operating System and explain the significance of privileged instructions in this context. (4 Marks)
2. Name and define three distinct types of malicious software (malware) (3 Marks)
3. Explain the difference between a Denial of Service (DoS) attack and a Distributed Denial of Service (DDoS) attack(3 Marks)

Question (6) Match the networking concept A with its B.(10 Marks): -

Column A	Column B
1. Octet	**A.** A cloud offering where the provider supplies IT managers access to physical network infrastructure, storage, and virtualized network resources.
2. IaaS (Infrastructure as a Service)	**B.** An operating system structure that moves functions from the kernel into user space, with communication between modules handled by message passing.
3. Microkernel	**C.** An 8-bit grouping, one of four sections comprising a 32-bit logical network address, commonly displayed in dotted-decimal notation.
4. Vishing	**D.** A network cable type composed of an inner wire surrounded by an insulator and a tubular conducting shield, providing improved protection against EMI.
5. Coaxial Cable	**E.** A newer form of social engineering that utilizes Voice over IP (VoIP) technology to deceive victims into releasing confidential information.
	F. A utility used to check the complete path a message takes, displaying each intermediary network device and the time taken for the round trip.

End of Questions

ACADEMYPT

CamScanner